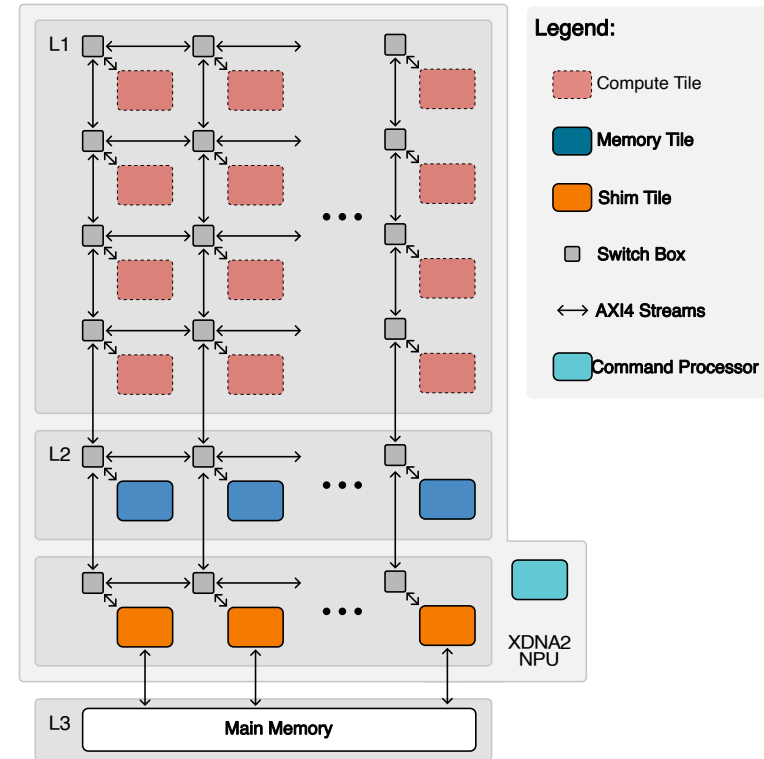




XDNA Data Movement

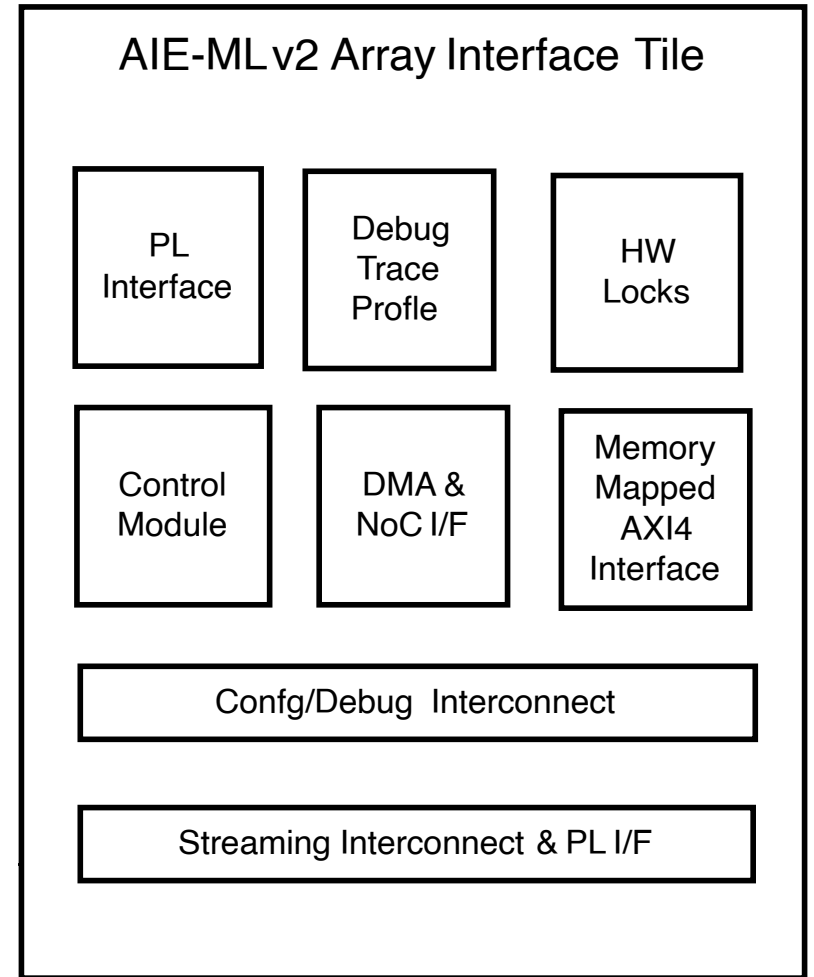
Tamino Steinert

Data Movement Hardware



Shim Tile

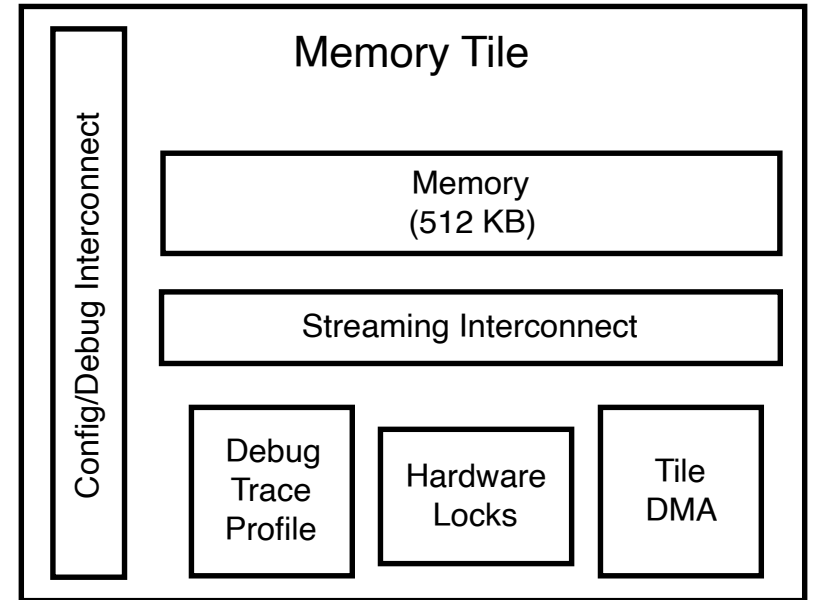
- Up to 57.6 GB/s bandwidth to the device (at a 1.8 GHz clock frequency)
- Up to 43.2 GB/s bandwidth from the device (at a 1.8 GHz clock frequency)
- Two DMA input streams and two DMA output streams
- 16 hardware locks (6-bit each) for synchronization
- 16 buffer descriptors (BDs)



Shim tile overview.
(AMD — AIE-ML v2 Architecture Manual — AM027)

Memory Tile

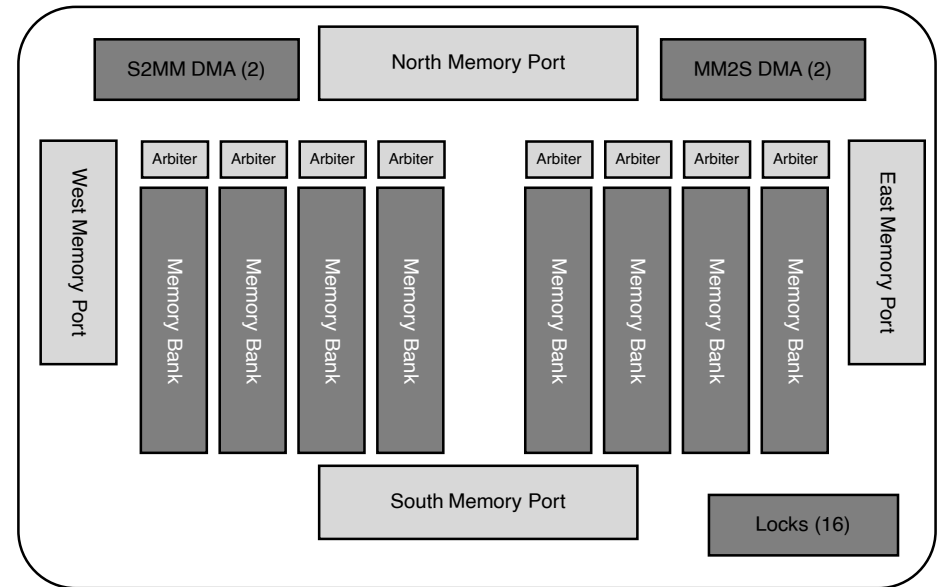
- 512 KiB data memory (L2 scratchpad memory)
- Six DMA input streams and six DMA output streams
- DMA streams read/write 64-bit data per cycle
- DMA controller supports 4D tensor address generation
- 64 hardware locks (6-bit each) for synchronization
- 48 BDs



Memory tile overview.
(adapted from AMD — AIE-ML v2 Architecture Manual — AM027)

Compute Tile Memory Module

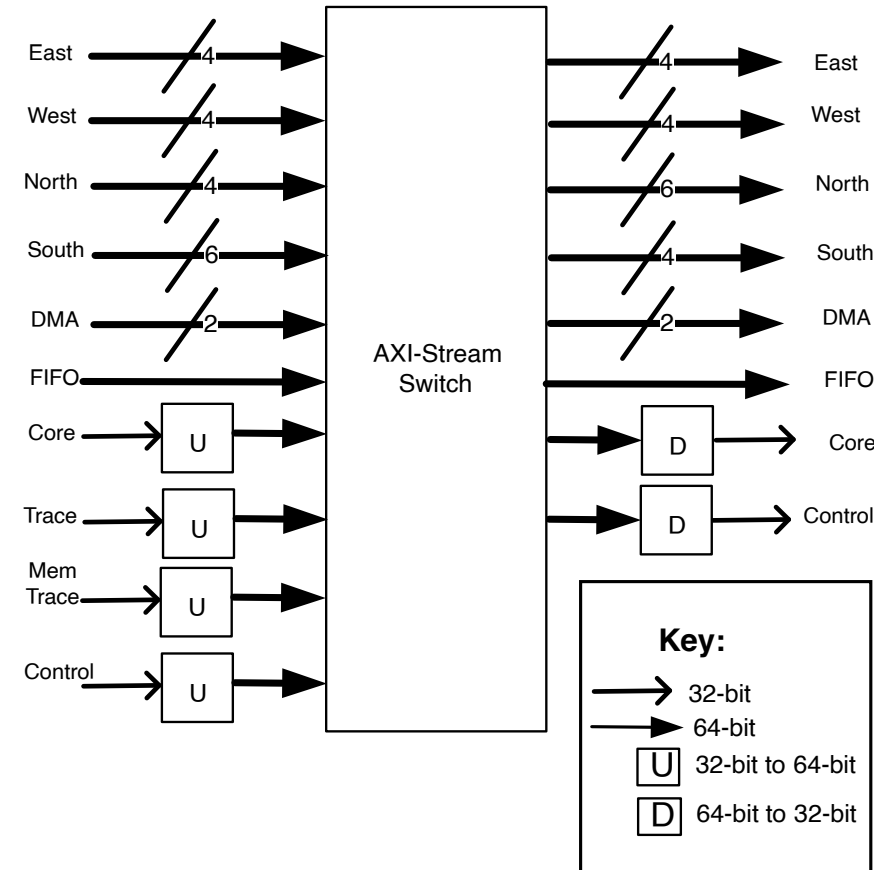
- 64 KiB data memory (L1 scratchpad memory)
- Data memory is organized into four 512-bit-wide banks (programmer's perspective)
- Two DMA input streams and two DMA output streams
- DMA streams read/write 64-bit data per cycle
- DMA controller supports 4D tensor address generation
- 16 hardware locks (6-bit each) for synchronization



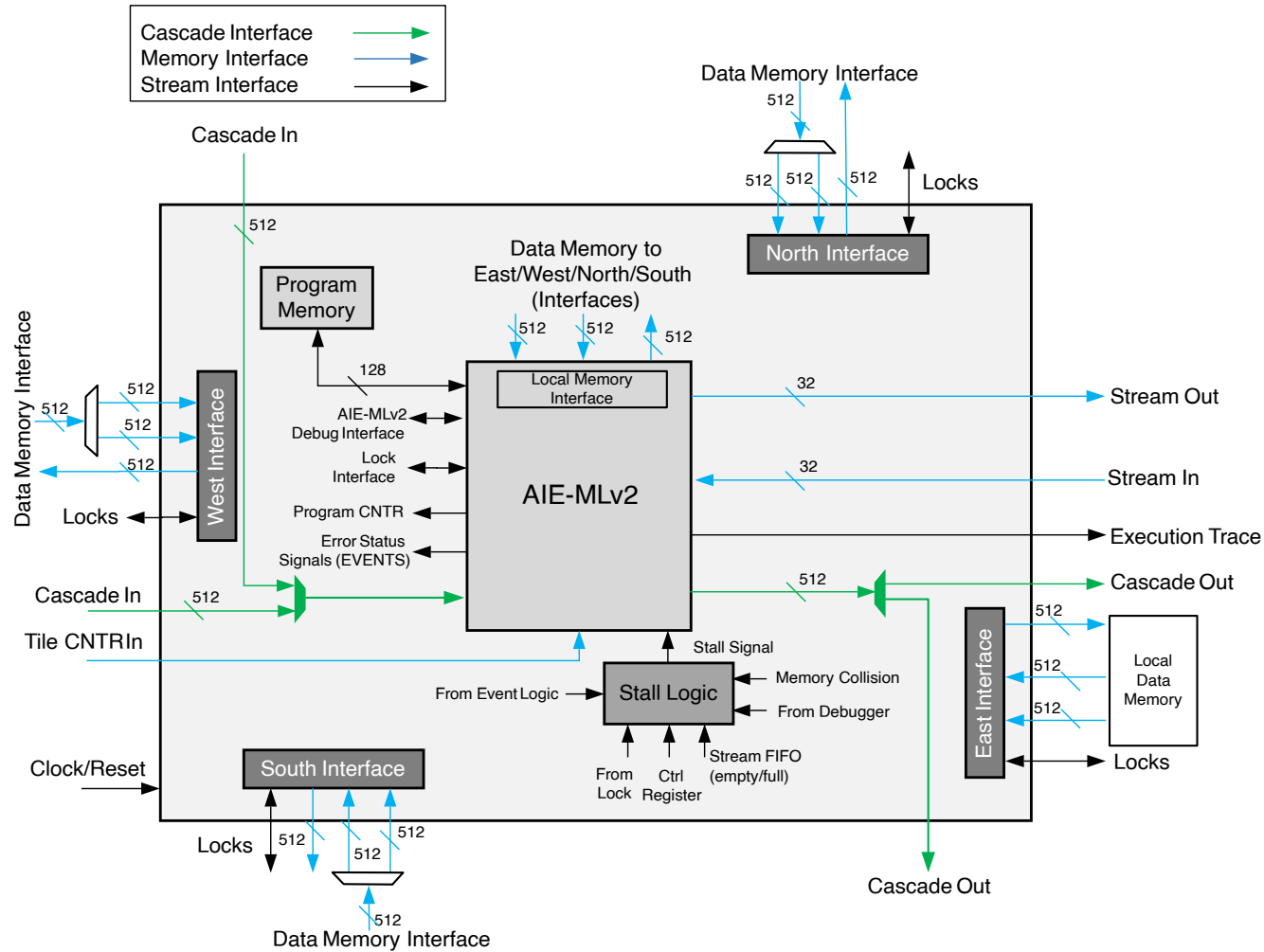
Compute tile memory module.
(AMD — AIE-ML v2 Architecture Manual — AM027)

Switch Box

- Connects tiles in a 2D layout, e.g., *North* ports (out) connect to the *South* ports (in) of the switch box above.
- Can connect ports internally, e.g., **north:1** (in) can connect to **west:2** (out).
- Can broadcast data by connecting one input port to multiple output ports.



Switch box overview (from compute tile).
(AMD — AIE-ML v2 Architecture Manual — AM027)

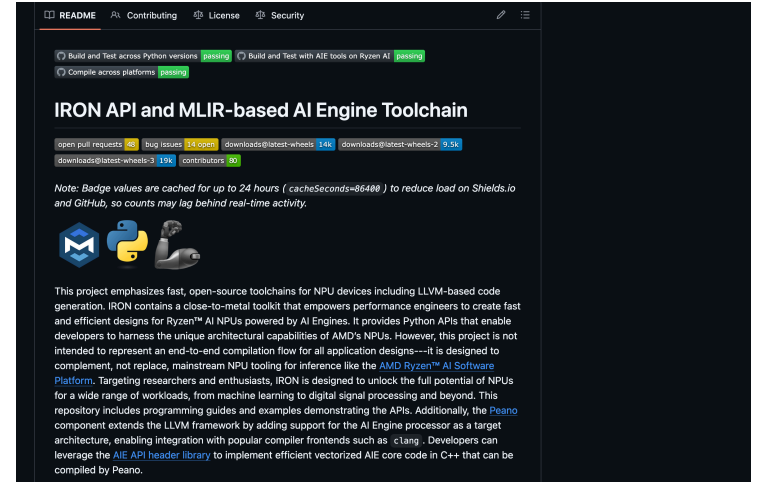


Compute tile interfaces.
(AMD — AIE-ML v2 Architecture Manual — AM027)

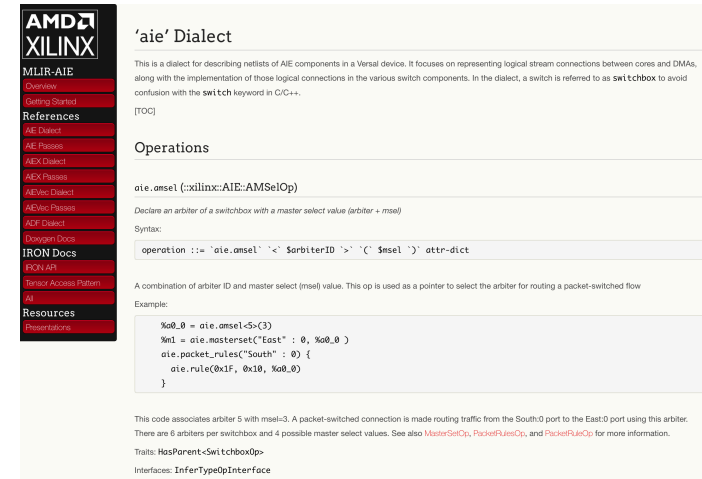
Programming Data Movement

MLIR-AIE

- XDNA has two MLIR (Multi-Level Intermediate Representation) dialects, i.e., **aie** and **aie.x**.
- **aie.x** contains *experimental* features that are promoted to **aie** once they mature.
- The highest level of abstraction uses FIFO queues, which are linked together.
- In FIFO queues, data transformations are described with sizes, strides, and offsets.
- Data can be repeated by setting a stride to zero and the corresponding size to 2 or larger (only on the outer-most, i.e., left-most, dimension).
- Broadcasts are done by linking one input FIFO queue to multiple output FIFO queues.



AMD — MLIR-AIE repository



AMD — MLIR-AIE dialects